

AP Calculus AB Practice Test

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Unit 1: Limits and Continuity

- Evaluate the limit: $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - x - 2}$
A. 0 B. $\frac{4}{3}$ C. 1 D. The limit does not exist
- Find the equation of the horizontal asymptote for the rational function: $f(x) = \frac{3x^3 - 5x + 2}{1 - 2x^3}$
A. $y = 3$ B. $y = 0$ C. $y = -\frac{3}{2}$ D. No horizontal asymptote
- Given the piecewise function $f(x) = \begin{cases} kx + 3, & x \leq 3 \\ k^2x - 9, & x > 3 \end{cases}$ is continuous at $x = 3$, the possible values for the real number k are:
A. $k = 2$ or $k = -2$ B. $k = 2$ or $k = -1$ C. $k = -2$ or $k = 1$ D. $k = 3$ or $k = -1$
- Evaluate the limit: $\lim_{x \rightarrow 0} \frac{\sin(4x)}{3x}$
A. $\frac{4}{3}$ B. $\frac{3}{4}$ C. 0 D. 1
- For the function $f(x) = \frac{x-1}{|x-1|}$, which of the following statements about $\lim_{x \rightarrow 1} f(x)$ is true?
A. The limit is 1 B. The limit is -1 C. The limit is 0 D. The limit does not exist because the left and right limits are not equal
- At which of the following points does the function $f(x) = \frac{x+2}{x^2-4}$ have a vertical asymptote?
A. Only at $x = 2$ B. Only at $x = -2$ C. At $x = 2$ and $x = -2$ D. No vertical asymptote exists
- Intermediate Value Theorem on a closed interval. Let $f(x)$ be continuous on the closed interval $[1, 5]$, with $f(1) = -2$ and $f(5) = 6$. Which of the following values of M is NOT guaranteed by the theorem to have a value c in the interval $(1, 5)$ such that $f(c) = M$?
A. 0 B. 3 C. -1 D. 7

Unit 2: Differentiation: Definition and Fundamental Properties

- The limit expression $\lim_{h \rightarrow 0} \frac{(2+h)^4 - 16}{h}$ represents the derivative of which of the following functions at the specified point?
A. $f(x) = x^4$ at $x = 2$ B. $f(x) = x^4$ at $x = 16$
C. $f(x) = (x+2)^4$ at $x = 0$ D. $f(x) = 16$ at $x = 2$
- If $f(x) = x^3 - 2x$, find the instantaneous rate of change of the function at $x = 2$.

A. 4 B. 10 C. 12 D. 14

10. Which of the following statements about the relationship between differentiability and continuity is true?

- A. A continuous function is always differentiable everywhere. B. A discontinuous function can be differentiable at some points.
C. A differentiable function is always continuous. D. If a function is not differentiable at a point, it must be discontinuous at that point.

11. Let $f(x) = (3x^2 + 1)(2x - 5)$. Find its derivative $f'(x)$.

- A. $f'(x) = 18x^2 - 30x + 2$ B. $f'(x) = 6x$
C. $f'(x) = 12x^2 - 30x$ D. $f'(x) = 18x^2 - 30x$

12. Given $f(x) = \frac{x^2 + 3}{x - 1}$, find the value of $f'(2)$.

- A. -3 B. -1 C. 1 D. 3

13. Find the slope of the tangent line to the curve $y = 2 \sin x + \cos x$ at the point $(0, 1)$.

- A. 0 B. 1 C. 2 D. -1

14. Let $f(x) = \frac{4}{\sqrt{x}} - 3x^2$. Find $f'(1)$.

- A. -8 B. -4 C. -2 D. 2

15. The position of an object on the closed interval $[0, 3]$ is given by $s(t) = t^2 + 2t$. What is the average velocity of the object over this time interval?

- A. 5 B. 8 C. 15 D. 3

Unit 3: Differentiation: Composite, Implicit, and Inverse Functions

16. Let $f(x) = \ln(x^2 + 3x)$. Find $f'(x)$.
- A. $\frac{1}{x^2 + 3x}$ B. $\frac{2x + 3}{x^2 + 3x}$ C. $\frac{2x}{x^2 + 3x}$ D. $\frac{1}{2x + 3}$
17. Given $y = e^{\sin x}$, find $\frac{dy}{dx}$.
- A. $e^{\sin x}$ B. $\cos x \cdot e^{\sin x}$ C. $-\cos x \cdot e^{\sin x}$ D. $\sin x \cdot e^{\sin x - 1}$
18. For the implicit function determined by the equation $x^2 + xy + y^2 = 7$, the derivative $\frac{dy}{dx}$ at the point $(1, 2)$ is:
- A. $-\frac{4}{5}$ B. $-\frac{5}{4}$ C. $-\frac{2}{5}$ D. 1
19. Let $f(x) = \arctan(2x)$. Find $f'(x)$.
- A. $\frac{2}{1 + 4x^2}$ B. $\frac{1}{1 + 4x^2}$ C. $\frac{2}{1 + x^2}$ D. $\frac{1}{1 + 2x^2}$
20. Let $g(x)$ be the inverse function of $f(x)$. If $f(3) = 5$ and $f'(3) = 4$, then the value of $g'(5)$ is:
- A. $\frac{1}{5}$ B. $\frac{1}{3}$ C. $\frac{1}{4}$ D. 4
21. Let $y = \cos^3(2x)$. Find $\frac{dy}{dx}$.
- A. $-3 \cos^2(2x) \sin(2x)$ B. $-6 \cos^2(2x) \sin(2x)$
C. $6 \cos^2(2x) \sin(2x)$ D. $-6 \sin^2(2x)$
22. For the curve determined by the equation $x^2 - y^2 = 9$, find its second derivative $\frac{d^2y}{dx^2}$ at any point.
- A. $\frac{1}{y}$ B. $\frac{x^2}{y^3}$ C. $-\frac{9}{y^3}$ D. $\frac{9}{y^3}$

Unit 4: Contextual Applications of Differentiation

23. A particle moves along a straight line with its position function given by $s(t) = 2t^3 - 9t^2 + 12t - 4$ (for $t \geq 0$). During which of the following time intervals is the particle moving to the left (in the negative direction)?
- A. $(0, 1)$ B. $(1, 2)$ C. $(2, \infty)$ D. The particle never moves to the left
24. Following the previous question, what is the acceleration $a(2)$ of the particle at $t = 2$?
- A. 0 B. 3 C. 6 D. 12
25. (Related Rates) The side length of a square is increasing at a constant rate of 3 cm/s. What is the rate of increase of its area when the side length is 10 cm?
- A. 30 cm²/s B. 60 cm²/s C. 90 cm²/s D. 100 cm²/s

26. (Related Rates) A ladder 5 m long is leaning against a vertical wall. If the base of the ladder is pulled away from the wall at a speed of 1.5 m/s, what is the downward speed of the top of the ladder when the base is 3 m from the wall?
- A. 1.125 m/s B. 2.0 m/s C. 1.5 m/s D. 0.875 m/s
27. Use the linear approximation of the function $f(x) = \sqrt{x}$ at $x = 4$ to estimate the value of $\sqrt{4.2}$.
- A. 2.025 B. 2.05 C. 2.1 D. 2.0125
28. Evaluate the limit using L'Hôpital's Rule: $\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{\sin(2x)}$
- A. 0 B. 1 C. $\frac{2}{3}$ D. $\frac{3}{2}$
29. Evaluate the limit: $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x^2}$
- A. 0 B. 1 C. ∞ D. The limit does not exist
30. A cylindrical water tank has a constant radius and a base area of $25\pi \text{ m}^2$. Water is being pumped into the tank at a constant rate of $5 \text{ m}^3/\text{min}$. What is the rate at which the water level h is rising?
- A. $\frac{1}{5} \text{ m/min}$ B. $\frac{1}{5\pi} \text{ m/min}$ C. 5 m/min D. $25\pi \text{ m/min}$
31. Evaluate the limit: $\lim_{x \rightarrow 0} \frac{\sin(5x)}{x}$
- A. 0 B. 1 C. 5 D. The limit does not exist
32. Evaluate the limit: $\lim_{x \rightarrow \infty} \frac{e^x}{x^3}$
- A. 0 B. 1 C. $\frac{1}{3}$ D. ∞
33. Evaluate the limit: $\lim_{x \rightarrow 2} \frac{x^2 - 4x + 4}{x^3 - 8}$
- A. 0 B. $\frac{1}{12}$ C. $\frac{1}{4}$ D. The limit does not exist
34. Evaluate the limit: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$
- A. 0 B. $\frac{1}{2}$ C. 1 D. ∞
35. Evaluate the limit: $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$
- A. 0 B. 1 C. e D. The limit does not exist
36. Evaluate the limit: $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$
- A. 0 B. $\frac{1}{2}$ C. 1 D. ∞

37. Evaluate the limit: $\lim_{x \rightarrow \infty} \frac{\ln(x^5)}{x}$

- A. 0 B. 1 C. 5 D. ∞

38. Evaluate the limit: $\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$

- A. 1 B. 0 C. -1 D. The limit does not exist

39. Evaluate the limit: $\lim_{x \rightarrow 0} \frac{x \cos x}{\sin(2x)}$

- A. 0 B. $\frac{1}{2}$ C. 1 D. 2

40. Evaluate the limit: $\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 7}{4x^2 + 1}$

- A. 0 B. $\frac{3}{4}$ C. 3 D. ∞